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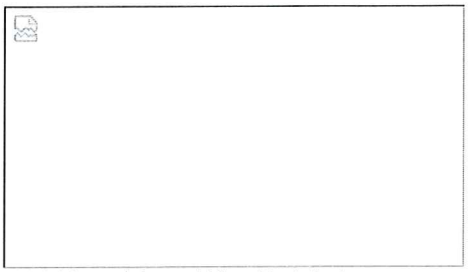
Bioterror

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Interviews with Biowarriors

Sam Kirkpatrick and Vince Kori

When Dr. Vlad Korsakov fled to the U.S. in 1992, he carried with him intimate knowledge of the Soviet Union's biological weapons program. Korsakov, who later changed his name to Vince Kory, had served as deputy chief of Biopreparat, the agency at the heart of the Soviet program.



Bill Kirkpatrick and Ken Kori, between them, have nearly 50 years of experience in biological weapons research.

To learn Soviet secrets, the U.S. government turned to longtime bioweaponeer Sam Kirkpatrick to debrief the defector in a series of clandestine meetings.

Here, prompted a NOVA producer, the two reminisce about their first meetings together. They also offer two insiders' views of the threat of bioterrorism today.

NOVA: Does the pervasiveness of biological weapons concern you? Are you worried about a threat either from a rogue state or from a terrorist organization?

Kori: My answer is yes. People don't realize biological weapons could be the most sophisticated weapons. Biological weapons could be used covertly. There are a lot of different deployment scenarios. There are a lot of different techniques to manufacture biological weapons. And a lot of different agents could be used in biological weapons.

One of the problems is biotechnology is moving fast. We see a lot of changes in biotechnology in general—in microbiology, genetic engineering. And all the developments will give more and more information about how to develop and manufacture sophisticated types of biological weapons. That worries me very, very much.

"My biggest concern now is a rogue country that supports state terrorism," notes Kirkpatrick.

Kirkpatrick: My take is very similar to Vince's. I don't think that Tom, Dick, and Harry terrorists, without significant training and experience in this arena, could develop an agent that would cause serious harm to this country. My biggest concern now is a rogue country that supports state terrorism and has the facilities to prepare, for example, a good dry powder of anthrax.

They could use that powder not in a large overt sense—large targets in outside environments—but they could certainly place that powder strategically in buildings, in subway systems. And they could cause serious harm. That's my biggest concern today.

My second biggest concern is what is happening to the scientists of the former Soviet Union who have the techniques and the knowledge base to manufacture a weapons agent. Where are they? Are they going to Iran? Are they going to Iraq, North Korea, perhaps? They could significantly improve the ability of the Iraqi program overnight with just a few changes in strains, just a few changes in production procedures, and transform Iraq into a very capable BW [biological weapons] country. These are where my concerns lie.

NOVA: What do you think, Ken? These are your colleagues.

Kori: You know, yesterday night I had a talk with [**Sergei Popov**], my friend and my coworker now. He was a department chief back in Russia working in the area of biological weapons. We discussed changes we can see now in Russia. And he was telling me about how it was easy to steal something from the BW facilities.

He was talking about groups of scientists with a desire to sell some products.United States, I realized I knew practically everything about the United States program.

"You can get this information just for the cost of a translator from Russian into English," says Kori.

"I just put my head down on the table ... and said, "Oh, my God. Oh, my, God," recalls Kirkpatrick.

Kirkpatrick: Right. And we knew absolutely nothing about yours. I never will forget when you started giving me the potential production figures for your various weaponized agents. If you recall, I just put my head down on the table where we were talking and said, "Oh, my God. Oh, my, God." It was a revelation that was just unbelievable to me.

Kori: Well, the Soviet Union had four major anthrax production facilities.

Kirkpatrick: Yeah.

Kori: One was located in the City of Kurgan. Another one was located in the City of Penza. One more in Sverdlovsk, which belonged to the Ministry of Defense, and one more relatively recently established in Stepnogorsk. And I became commander of this facility, the Stepnogorsk facility, in 1983 with the specific task to develop new anthrax biological weapons.

Kirkpatrick: How old were you, Ken, in 1983?

Kori: Thirty-two years old.

Kirkpatrick: That is a lot of responsibility for someone that young. You were doing the things that were necessary to weaponize agents, and you were very effective in doing that job.

Kori: But, you know, when I finished this work in 1986, I was a little disappointed because the production facility I ran at that time was not able to manufacture more than 300 tons of the anthrax during a 250-day period.

Kirkpatrick: [Laughter.] Oh, my goodness. Only 300 tons.

Kori: I was disappointed because I thought I would be able to reach something like 1,000 tons. But when I came to the United States, I realized that it was a little more than you were able to manufacture.

"You were out-producing us by a factor of 300-to-1," Kirkpatrick notes.

Kirkpatrick: Oh, my goodness. You were out-producing us by a factor of 300-to-1 in one plant.

NOVA: Ken, how did you feel about Bill Kirkpatrick when you met him? What made you think, this is a guy that I can talk to?

Kori: First, because I liked you. Of course, I didn't understand even a word you were saying because I didn't speak English.

Kirkpatrick: Yeah.

Kori: I didn't understand English, but it was clear to me you were quite a funny person in a good sense.

Kirkpatrick: Dallas humor, I think, is the expression.

Kori: And even at that time I was not absolutely sure that the United States had terminated the BW program.

Kirkpatrick: Yeah. Right.

Kori: I was trying to figure out whether or not you were a real guy who worked in this field. And when we started a discussion—it was not a very long discussion—but it started getting clear that you knew this issue firsthand.

Kirkpatrick: Well, you paid me a very high compliment in your book *Biohazard* where [you write] that you couldn't convince anyone about the procedures and capabilities [of the Soviet program], and you said that I was the first person who truly realized what that program was all about.

Regrets of bioweapons work

NOVA: When you look back on your career, do you have regrets about what you did? Do you have second thoughts about what you did in the moral sense?

Kori: Was it right for me to be involved in this business? I say no. Probably, I didn't have the right because of one important reason: Because I'm a physician.

Somebody with no medical background—a chemist or biologist who didn't pledge not to cause harm—I cannot accuse this person of wrongdoing. But with my specific case, I was a physician. I tried to escape—I tried not to be

involved in this business—but I was not able to do this.

NOVA: Bill, what about your take on the question?

Kirkpatrick still stresses, "We all felt that we were patriotic."

Kirkpatrick: Well, I've been posed this question many times, and I feel very ambivalent about providing a very cut-and-dried answer. I vacillate on this point. All I can say is, in retrospect, it probably was a good thing that President Nixon disestablished the offensive program in 1969.

But at the time, those of us who worked at Detrick [the U.S. Army's base for biological weapons research], we all felt that we were patriotic. We realized that we were in the midst of a very severe Cold War—with the United States on one side and the Soviet Union and, at that time, Red China on the other side. And we felt that it was absolutely the patriotic thing to do to develop biological warfare in case it was used against us. The policy of this country was always to have biological warfare available in a response, not first usage, but in response.

NOVA: So, in short, no regrets?

Kirkpatrick: The tempo of the times allowed me to work on offensive biological warfare agents without qualms. Without any qualms, yeah.

Learn more about the U.S. program in Bill Kirkpatrick's detailed interview. Or, hear from former Soviet researcher Sergei Popov, who developed novel biological weapons with genetic engineering.

kill cancer cells
= kill ~~all~~ any cells?

"Ethnic bioweapon"?